

Agenda

- ① What are tokens?
- ② Constants
- ③ Variables
- ④ Keywords
- ⑤ Comments

Tokens

- Tokens are also known as lexical elements
↓
शब्द
- The text of C/C++ program consists of tokens and white space.
- The token is the smallest element of a C/C++ program that is meaningful to the compiler.
- Parser is a program whose job is to separate these tokens.

Tokens

· Parser recognizes these kinds of tokens:

- Constants (Literals)
- Identifiers (variables, functions ...)
- Keywords
- operators
- Punctuators

White Space

- Tokens are usually separated by white space, which can be one or more
 - Blanks
 - Tabs
 - New lines
 - Form feed
 - Comments

Comments

- Comments are text ignored by compiler
- Comments are useful for documentation of your code and useful for programmers.

There are two types of comments

① Single line comment

// text here

② Multiline comment

```
/*  
  text line 1  
  text line 2  
  ....  
*/
```

Sample Program in C

```
int main( )  
{
```

```
    int a,b,c; //variables
```

```
    printf("Enter two numbers");
```

```
    scanf("%d %d", &a, &b);
```

```
    c=a+b;
```

```
    printf("sum is %d", c);
```

```
    /*main function body ends here */  
}
```

Operators

Punctuators

Constants

Identifiers

Keywords

Comments

Information = Constants = data

primary

Integer 35 0 -2 -421

Real 3.5 -0.05 2.0

Character 'A' 'a' '+' '3'
' ' '@'

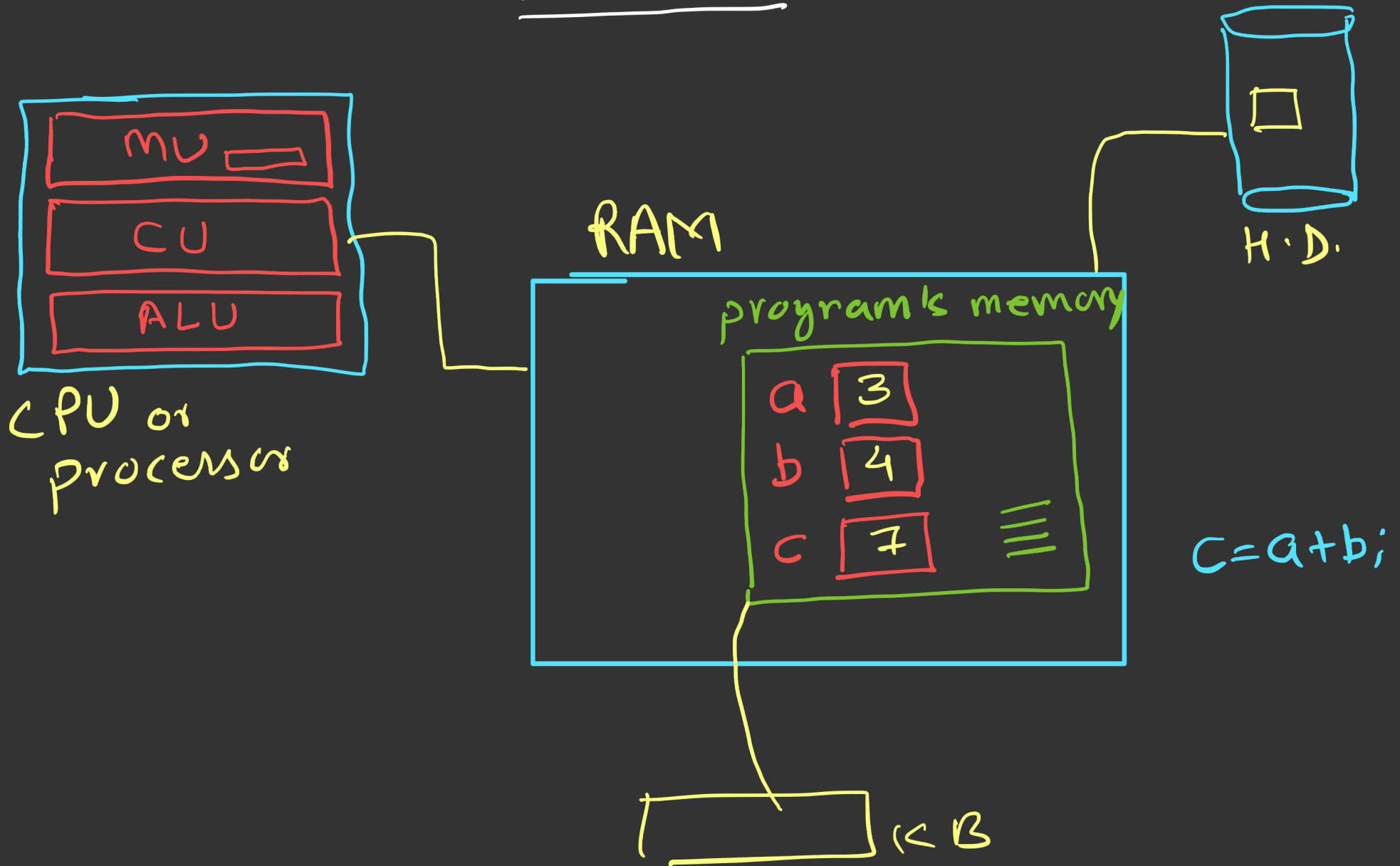
'3.4' '-5'

'\n'

Secondary

- pointers
- array
- string → "BHOPAL"
- structure
- union
- enumerator

Variables



Variables

- Variable is a name of memory location where we store program's data during execution of program.
- variable name is any combination of alphabets (a to z or A to Z), digits (0 to 9) and underscore (_)
- No other symbol is allowed.
- valid variable name cannot start from a digit.
- variable name cannot be a keyword.

a_2 ✓

$a_{b_2_3}$ ✓

x_1 ✗

s_i ✗

x_y ✗

$_A$ ✓

$2x$ ✗

C is a case sensitive language

Keywords in C predefined words / Reserved words

auto	double	int	struct
break	else	long	switch
case	enum	register	typedef
char	extern	return	union
const	float	short	unsigned
continue	for	signed	void
default	goto	sizeof	volatile
do	if	static	while

-Bool	-Alignas
-Complex	-Alignof
-Imaginary	-Atomic
inline	-Generic
restrict	-Noreturn
	-Static_assert
	-Thread_local

C89
32 keywords

C99
+5

C11
+7

Keywords in C++20

alignas	co_return	if	static
alignof	co_yield	inline	static_assert
asm	decltype	int	static_cast
auto	default	long	struct
bool	delete	mutable	switch
break	do	namespace	template
case	double	new	this
catch	dynamic_cast	noexcept	thread-local
char	else	nullptr	throw
char8_t	enum	operator	true
char16_t	explicit	private	try
char32_t	export	protected	typedef
class	extern	public	typeid
concept	false	register	typename
const	float	reinterpret_cast	union
constexpr	for	requires	unsigned
constexpr	friend	return	using
constinit	goto	short	virtual
const_cast		signed	void
continue		sizeof	volatile
co_await			wchar_t
			while